

Introduction	1
Definitions	1.1
Context picture	1.2
Historical perspective	1.3
Basic concepts	1.4-1.5
Fundamental properties of an encryption algorithm	1.6
Model of eve	1.7
Goals of a system designer	1.8
Background mathematics: Finite Field Theory	2
Properties of the following	2.1-2.3
Group	2.1
Abelian Ring	2.1
Ring	2.2
Commutative Ring	2.2
Integral Domain	2.2
Field	2.2
Modular arithmetic	2.4
Greatest common divisor (gcd)	2.5
Finite field properties	2.6
Euclidean algorithm for gcd	2.7
Polynomial arithmetic	2.8-2.10
Convert polynomials into a field	2.11-2.12
Addition	2.13
Multiplication	2.13
Division	2.14
Number theory	3
Prime numbers	3.1
Fundamental theorem of arithmetic	3.1
Fermat's little theorem	3.3

Euler's totient function: Euler's Theorem	3.4
Finding prime numbers	3.5
Miller-Rabin algorithm	3.5-3.7
Quantity of probable primes	3.8
Chinese remainder theorem	3.9-3.13
Power of an integer modulo n	3.14-3.15
Discrete logarithms	3.16
Complexity of discrete log	3.17
Stream Ciphers	4
One time pad (OTP)	4.1-4.2
One time pad limitations	4.3
Properties of stream ciphers	4.3
RC4	4.4-4.5
Salsa20	4.6
Cryptography – Classical Encryption Techniques	5
Substitution cipher	5.1-5.11
Caesar Cipher	5.2-5.3
Monoalphabetic Cipher	5.4-5.6
Playfair Cipher	5.7-5.8
Hill Cipher	5.9-5.11
Polyalphabetic substitution ciphers	5.12-5.20
Vigenere cipher	5.13-5.16
Autokey cipher	5.17-5.18
Vernam cipher	5.19
One time pad (OTP)	5.20
Rotor machine	5.21
Transposition ciphers	5.22
Entering the Modern Era of Symmetric Ciphers	6
Claude Shannon	6.1-6.9

Substitution-Permutation Networks (SPN)	6.2-6.5
Design decisions	6.4
Keying	6.4
Avalanche effect	6.5
Feistel Network	6.6-6.9
Design decisions	6.7
Keying	6.8
Invertibility without invertibility	6.9
Data Encryption standard (DES)	6.10-6.26
Overall structure	6.11-6.12
Round key generation algorithm	6.13
Known design criteria for DES	6.14
Security of DES	6.15-6.23
Differential and Linear Cryptanalysis	6.16-6.17
Time-memory trade-off attack	6.18-6.20
Increasing the key size of DES?	6.21
Double DES (2DES)	6.22-6.23
Tripple DES (3DES)	6.24-6.25
Advanced Encryption Standard (AES) algorithm	7
S-box construction	7.2
Modes of operations for block ciphers	7.3-7.6
Electronic Codebook (ECB) mode	7.3
Cipher Block Chaining (CBC) mode	7.3
Cipher Feedback (CFB) mode	7.4
Output feedback (OFB) mode	7.4
Counter (CTR) mode	7.5
Galois Counter Mode (GCM)	7.6
Symmetric Key Derivation	8
Random number generation (PSRG)	8.1-8.3

Linear Congruential Generator	8.3
Cryptographically generated random numbers	8.4-8.7
Counter (CTR) mode	8.4
ANSI X9.17 PRNG	8.5
PGP PRNG	8.6
Blum Blum Shub (BBS) Generator	8.7
Symmetric Key Distribution	8.8-8.10
Kerberos	8.9-8.10
Asymmetric (Public Key Encryption)	9
Diffie-Hellman	9.1
Terminology	9.2
Requirements for public key cryptography	9.3
One-way function	9.4
Trap-door one-way function	9.4
Cryptanalysis in public key algorithms	9.4
RSA algorithm	9.5-9.9
Considerations for computation and efficient operation	9.7-9.8
Security of RSA	9.9
Elliptic Curve Cryptography (ECC)	10
Elliptic curves over Real Numbers	10.1-10.3
Addition	10.2-10.3
Elliptic curves over Finite Fields	10.3-10.4
Elliptic curves over $GF(2^m)$	10.5
ECC encryption/decryption	10.6
Generators over $GF(2^m)$	11
Polynomial example	11.2
Elliptic curve example	11.3-11.4
Asymmetric Key Derivation	12
Finding prime numbers	12.1

Key distribution	12.2-12.3
Announcement	12.2
Directory	12.2
Online Authority	12.2
Certificates	12.3
Diffie-Hellman (D-H) protocol	12.3
Special Techniques in Confidentiality	13
Identity based encryption	13.1
Attribute Based Encryption	13.1
Fully Homomorphic Encryption	13.1
Broadcast Encryption	13.2
Oblivious Transfer	13.2
Integrity	14
Hash function	14.1-14.9
Requirements of a cryptographic hash function	14.2
Birthday paradox	14.3-14.4
Security of hash from birthday paradox	14.5
General structure	14.6-14.7
Secure hash algorithm (SHA)	14.8-14.9
Message authentication	14.10
Message Authentication Codes (MAC)	14.11-
Benefits of using MAC	14.11
Size of MAC, key, and message	14.12
Requirements of a MAC function	14.13
MAC construction	14.14
HMAC (MAC construction with hashing)	14.15
Authenticity	15
Digital signatures	15.1-15.5
Requirements for a digital signature	15.2

Arbitrated signatures	15.3
Direct signatures	15.4
Contrast with public key algorithms	15.5
RSA for digital signatures	
Weakness to no-message attack	15.7
Forgeries	15.8
Message size limitation	15.9
Solution hash & sign	15.9-15.11
Finding collision for H	15.10
Size of hash required	15.11
Digital Signature standard (DSS-> DSA)	15.12-15.17
Algorithm	15.13-15.14
Computational efficiency	15.15
Uniqueness of k	15.16
Cannot be used for encryption proof	15.17
Smaller key signatures than RSA	15.17
Special Types of Signatures	16
Blind signatures	16.1
Malleable Signatures	16.2
Authentication protocols (MACs or Signatures)	16.3-16.5
One-way authentication	16.3
Two-way authentication	16.4
Three-way authentication	16.5
Authentication in conjunction with other types of security	16.6-16.8
Authenticated encryption (AE)	16.6
Authenticated key exchange (AKE)	16.7
Authenticated version of DH (STS protocol)	16.8
Cryptographic strength	17
Defining strength	17.1

Attacker models	17.2-17.3
Passive	17.2
Active	17.2
Man in the middle (MitM)	17.2
Honest but Curious	17.2
Fail-stop	17.3
Malicious	17.3
Byzantine	17.3
Types of cryptographic problems	17.4
Computational complexity theory	17.4-17.5
Example	17.5
Mathematical models of computation (Turing machines TM)	17.5
Big-O/Little-O notation	17.6
Classes for TM models	17.7
Deterministic	17.7
Linear (DTIME)	17.7
Polynomial (P)	17.7
Exponential (EXPTIME)	17.7
Nondeterministic	17.7
Linear (N time)	17.7
Polynomial (NP)	17.7
Exponential (NEXPTIME)	17.7
Reduction	17.8
Hard problem	17.8
Complete Problem	17.8
Choosing the attacker model	17.9-17.11
General theory	17.9
Approach 1: polynomial bound attacker	17.10
Approach 2: no computational limitations	17.11

Proof of security	17.12-17.14
Step1: Theoretical model of the environment	17.12
Step2: Assumption of difficulty for mathematical computations	17.13
Step 3: Reduction	17.14
Information Theoretic Security	18
Commitment scheme	18.1
Security of a commitment scheme	18.2
Types of adversaries	18.2
Unconditional hiding vs. binding	18.2
Counterexamples using encryption and hashing	18.3
Pedersen commitment scheme	18.4-18.5
Zero knowledge proof of knowledge (ZKPoK) example	18.6
ZKPoK for a Pedersen commitment	18.7
ZKPoK with a revealed value	18.8
Quantum Computers and the Implication for Cryptography	19
Overview	19.1
Quantum algorithms	19.2
Grover's algorithm	19.2
Brassard, Hayer & Tapp	19.2
Shor's algorithm	19.2
Quantum problems for asymmetric cryptography	19.3-19.4
Lattices	19.3
Error correcting codes	19.3
Multivariate polynomial (MVP)	19.3
Hashbased	19.3
Isogenies	19.4
Other types	19.4
Quantum security Types	19.5
Post-Quantum Cryptography (PQC)	19.5

Quantum Key Distribution (QKD)	19.5
NIST standardization for PQC algorithms	19.6-19.7
To be standardized	19.6
CRYSTALS-KYBER	19-.6
CRYSTALS-DILITHIUM	19.6
FALCON	19.6
SPHINCS+	19.6
Continued to the next (4 th) round	19.7
BIKE, HQC	19.7
SIKE	19.7
Classic McEliece	19.7
Security levels for post-quantum candidates (NIST)	19.8-19.9